



Power Query

Data Transformation with Excel & Power BI

Chapter 1

Power Query Editor — Overview

Welcome to the world of Power Query! This chapter introduces you to one of the most powerful data tools available in Microsoft Excel and Power BI. By the end of this chapter, you will clearly understand what Power Query is, why it exists, and how to navigate its interface.

1.1 What is Power Query?

Power Query

A built-in data connection and transformation tool available in Microsoft Excel and Power BI. It allows you to import data from various sources, clean it, reshape it, and load it — without writing any code.

Think of Power Query as your personal data assistant. Whenever you receive raw, messy data — from a spreadsheet, a database, a website, or a CSV file — Power Query helps you clean and organize it automatically, every single time you refresh.

Real-World Analogy

Imagine you receive a monthly sales report from three different regional managers. Each report uses different date formats, different column names, and some rows have missing values.

Without Power Query: You manually fix each file every month — a process that takes hours and is prone to errors.

With Power Query: You set up the cleaning steps once. Next month, you simply click Refresh, and all three files are cleaned, combined, and ready in seconds.

1.2 Where Is Power Query Available?

Power Query is built into two widely used Microsoft products:

Product	How to Access Power Query
Microsoft Excel 2016 and later	Data tab → Get Data → Launch Power Query Editor
Microsoft Excel 365	Data tab → Get Data → Launch Power Query Editor
Power BI Desktop	Home tab → Transform Data → Opens Power Query Editor



Note

This textbook uses Microsoft Excel as the primary platform for demonstrations. All concepts apply equally to Power BI. Screenshots will show Excel's interface — the Power Query Editor looks identical in both products.

1.3 The Power Query Editor — Interface Overview

When you open Power Query, a separate window called the Power Query Editor opens. This is where all your data transformation work happens. Understanding each part of this interface is the first step to working efficiently.

The Power Query Editor consists of five main areas:

N o.	Area	What It Does
1	Ribbon / Toolbar	Contains all the commands and tools organized into tabs — Home, Transform, Add Column, View.
2	Query Pane (Left Panel)	Lists all queries (data connections) you have created. You can switch between multiple queries here.
3	Data Preview (Centre)	Shows a preview of your data — the rows and columns as they appear after your transformations.
4	Applied Steps (Right Panel)	Records every transformation you apply — like a recipe. You can review, edit, or delete any step.
5	Formula Bar	Displays the M formula (the code behind each step). You can view and manually edit it if needed.

1.3.1 The Ribbon Tabs Explained

Let us look at each Ribbon tab and what you will find there:

Tab	Key Commands Available
Home	Close & Load, Refresh, Remove Rows, Keep Rows, Sort, Group By, Combine Queries
Transform	Change Data Type, Rename Column, Split Column, Pivot, Unpivot, Fill, Extract, Replace Values
Add Column	Add Custom Column, Conditional Column, Index Column, Duplicate Column, Column From Examples
View	Show/Hide Formula Bar, Query Settings, Column Quality, Column Distribution, Column Profile

1.3.2 Query Settings Panel (Applied Steps)

On the right side of the Power Query Editor is the Query Settings panel. The most important part of this panel is the Applied Steps list.

Applied Steps: Every time you perform an action — such as removing a column, changing a data type, or filtering a row — Power Query records it as a step. These steps are stored in order, like a recipe.

Pro Tip

You can click on any Applied Step to see exactly what your data looked like at that point. This makes it very easy to go back and fix a mistake — simply delete the incorrect step or edit it.

1.4 Importing Data into Power Query

Before you can clean or transform any data, you need to import it into Power Query. Power Query can connect to many different data sources.

Data Source	Examples
Files	Excel workbooks (.xlsx), CSV files (.csv), Text files (.txt), JSON, XML
Databases	SQL Server, MySQL, Oracle, Access, PostgreSQL
Online Services	SharePoint, Microsoft 365, Google Analytics, Salesforce
Web	Any table published on a public website (using URL)
Other	Blank Query (manual input), OData feeds, Azure services

To import a CSV file (most common for beginners):

- Step 1: Go to the Data tab in Excel.
- Step 2: Click Get Data → From File → From Text/CSV.
- Step 3: Browse and select your file. A preview window appears.
- Step 4: Click Transform Data. The Power Query Editor opens.



Chapter 2

Data Cleaning — Remove, Rename & Change Data Types

Raw data is rarely perfect. It often contains extra columns you do not need, rows with errors or blanks, column names that are unclear, and values stored in the wrong format. This chapter teaches you how to clean data systematically using Power Query.

2.1 Removing Rows

Sometimes your dataset contains rows that are not useful — blank rows, error rows, duplicate rows, or rows that do not meet your criteria. Power Query provides dedicated commands to remove them cleanly.

2.1.1 Remove Top / Bottom Rows

Use this when your raw data has header information at the top (like a company name or report title) that is not part of the actual data.

How to do it:

- Go to the Home tab in the Ribbon.
- Click Remove Rows → Remove Top Rows.
- Enter the number of rows to remove (e.g., 2). Click OK.

Example

You have a monthly sales CSV exported from your company's ERP system. The first two rows contain:
Row 1: 'ACME Corporation — Monthly Sales Report'
Row 2: 'Generated on: 01-Apr-2025'

These are not data rows. Remove the top 2 rows to get to the actual column headers and data.

2.1.2 Remove Blank Rows

A blank row is a row where all cells are empty. These often appear in exported files and can cause errors in calculations.

- Go to Home tab → Remove Rows → Remove Blank Rows.
- Power Query automatically identifies and removes every row that has no values.

2.1.3 Remove Duplicate Rows

Duplicate rows appear when the same record is entered more than once. For example, a customer might appear twice in an order list.

- Select the column(s) that define uniqueness (e.g., Order ID, Customer ID).
- Go to Home tab → Remove Rows → Remove Duplicates.
- Power Query keeps only the first occurrence and removes all subsequent duplicates.

**Note**

If you click Remove Duplicates without selecting a specific column, Power Query checks all columns together. A row is only considered a duplicate if every single column matches another row exactly.

2.1.4 Remove Rows with Errors

When data is imported, certain values may cause errors — for example, dividing by zero or importing incompatible data types. These show up as Error in the data preview.

- Go to Home tab → Remove Rows → Remove Errors.
- All rows containing an error in any column will be deleted.

2.2 Removing Columns

Your dataset may contain columns you do not need for your analysis — perhaps internal codes, system fields, or irrelevant metadata. Removing them keeps your data clean and reduces clutter.

2.2.1 Remove Specific Columns

- Click on the column header you want to remove (hold Ctrl to select multiple columns).
- Right-click → Remove Columns. OR Go to Home tab → Remove Columns → Remove Columns.

2.2.2 Remove Other Columns (Keep Only Selected)

This is the reverse of removing specific columns. If you only need 3 columns from a 20-column dataset, it is faster to select the 3 you want and remove everything else.

- Select the columns you want to keep (hold Ctrl to select multiple).
- Right-click → Remove Other Columns.
- Only the selected columns remain. All others are deleted.

**Pro Tip**

Use Remove Other Columns when you need only a few columns from a wide dataset. It is faster than removing each unwanted column one by one.

2.3 Renaming Columns

Good column names make your data easier to understand. Imported data often has names like 'Col1', 'Unnamed: 3', or overly long system-generated labels. Renaming them makes your reports professional and readable.

How to rename a column:

- Method 1: Double-click the column header. Type the new name. Press Enter.
- Method 2: Right-click the column header → Rename. Type the new name. Press Enter.
- Method 3: Select the column → Transform tab → Rename.

Example

Your imported file has these column names: 'cust_id', 'f_nm', 'l_nm', 'phn', 'amt_paid'

After renaming them to: 'Customer ID', 'First Name', 'Last Name', 'Phone Number', 'Amount Paid'

Your report becomes instantly readable to anyone — including non-technical stakeholders.

2.4 Changing Data Types

A data type tells Power Query what kind of value a column contains — is it a number, a date, or text? Assigning the correct data type is one of the most important steps in data cleaning.

If a column has the wrong data type, calculations will fail or give wrong results. For example, a column containing salaries stored as Text cannot be summed — it must be changed to a Number type first.

Data Type	Icon in Power Query	Use When the Column Contains...
Whole Number	123	Integers — quantities, IDs, counts (e.g., 42, 1000)
Decimal Number	1.2	Numbers with decimal places (e.g., 99.99, 3.14)
Text	ABC	Names, descriptions, codes, IDs as text (e.g., 'Alice', 'EMP-001')
Date		Dates only — no time (e.g., 01/04/2025)
Date/Time		Both date and time (e.g., 01/04/2025 14:30:00)
True/False	✓ ✗	Boolean values — Yes/No, True/False flags

Currency	\$	Monetary values (e.g., 1,499.00)
Percentage	%	Decimal values representing percentages (e.g., 0.15 = 15%)

How to change a data type:

- Click the data type icon to the left of the column name (the small icon like '123' or 'ABC').
- A dropdown list of available data types appears. Select the correct type.
- Alternatively: Right-click the column header → Change Type → select the type.

 Common Mistake

Dates imported from CSV files are often treated as Text by Power Query. If you try to filter by date range or calculate the number of days between dates, it will not work until you change the type to Date.

Always verify the data type of every column before loading data into your report.



Chapter 3

Handling Missing Values

In the real world, datasets are almost never complete. A customer record might be missing a phone number. A sales entry might have no discount value. A survey might have unanswered questions. Missing values are represented as null in Power Query.

This chapter teaches you how to detect, handle, and replace these missing values correctly.

3.1 What Is null?

null

A special placeholder in Power Query that represents a missing or unknown value. It is not zero, not an empty string — it simply means 'no value exists here'.



Analogy

Think of null like a blank field on a paper form — the space exists, but nothing was written in it. It is different from writing '0' (which means the number zero) or 'N/A' (which is a text value).

3.2 Detecting Missing Values

Power Query has a built-in profiling feature that shows you a summary of missing values for each column.

How to enable Column Profiling:

- Go to the View tab in the Ribbon.
- Turn on Column Quality, Column Distribution, and Column Profile.

Once enabled, you will see a quality bar below each column header showing:

- Valid: percentage of rows with valid values (shown in green).
- Error: percentage of rows with error values (shown in red).
- Empty: percentage of rows with null/blank values (shown in grey).

3.3 Removing Rows with Missing Values

If you do not need rows where a key field is missing, you can remove them.

- Click the filter dropdown arrow on the column header.
- Uncheck (null) from the filter list. Click OK.
- All rows where that column is null will be removed.

**Note**

You can also go to Home tab → Remove Rows → Remove Blank Rows to remove rows where all columns are empty. For removing rows based on null in a specific column, use the filter method described above.

3.4 Replacing Missing Values

In many cases, you do not want to remove a row — you want to fill in a sensible value in place of null. This is called replacing or imputing missing values.

3.4.1 Replace with a Fixed Value

This is the most straightforward method. You choose a specific value to replace every null in a column.

- Right-click the column header → Replace Values.
- In the Value to Find field, leave it blank (this targets null values).
- In the Replace With field, enter the value you want (e.g., 0, 'Unknown', 'N/A').
- Click OK.

**Example**

A product catalogue has a 'Discount %' column. Some products have no discount, so the field is null.

You replace all nulls in 'Discount %' with 0, because a null discount is effectively a 0% discount. This allows you to calculate final prices without errors.

3.4.2 Fill Down / Fill Up

Fill Down and Fill Up are useful when a value is stated once at the top of a group of rows, and all the rows below (or above) it should inherit that same value.

This is common in reports that have merged cells or grouped data when exported:

Region	Salesperson	Sales Amount
North	Alice	50,000
null	Bob	42,000
null	Charlie	38,000
South	Diana	61,000

null	Edward	55,000
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Figure 3.5 — Example of null values that should be filled down

The Region column says 'North' once but belongs to three salespeople. Fill Down fixes this:

- Select the Region column.
- Go to Transform tab → Fill → Down.
- Power Query copies each non-null value downward until it reaches the next non-null value.



Chapter 4

Splitting & Merging Columns

Data often arrives in a format that needs reshaping. Full names may be in one column when you need first and last name separately. Or city and postal code may be combined in one field. This chapter covers how to split one column into multiple columns and how to merge multiple columns into one.

4.1 Splitting Columns

Splitting a column divides its content into two or more new columns based on a rule you define.

4.1.1 Split by Delimiter

A delimiter is a character that separates values within a cell — such as a comma, hyphen, space, or slash.

Example

Column 'Full Name' contains values like: 'Alice Sharma', 'Bob Mehta', 'Charlie Patel'

You want to split this into two columns: 'First Name' and 'Last Name'.
The delimiter is a space character ().

Steps:

- Select the column you want to split.
- Go to Transform tab → Split Column → By Delimiter.
- Choose the delimiter from the dropdown (comma, tab, space, or enter a custom one).
- Choose split position: At each occurrence, At the left-most delimiter, or At the right-most delimiter.
- Click OK. Power Query creates new columns with names like ColumnName.1 and ColumnName.2.
- Rename the new columns appropriately.

4.1.2 Split by Number of Characters

Use this when the split position is based on a fixed number of characters rather than a delimiter.

Example

You have a column 'Product Code' with values like: 'CAT001', 'CAT002', 'ELE043'
The first 3 characters represent the category ('CAT', 'ELE').

The last 3 characters represent the product number ('001', '043').

Split by Number of Characters (3 characters, once, from the left) gives two columns.

- Transform tab → Split Column → By Number of Characters.
- Enter the number of characters. Choose to split from left or right. Click OK.

4.1.3 Split by Position

Split by Position lets you specify exact character positions where the column should be divided. This is useful for fixed-width data — common in legacy financial and banking exports.

4.2 Merging Columns

Merging (also called combining) columns joins the values from two or more columns into a single column, with an optional separator between them.

Example

You have two separate columns: 'First Name' = 'Alice' and 'Last Name' = 'Sharma'.
You want to create a single column 'Full Name' = 'Alice Sharma'.

Steps to merge columns:

- Hold Ctrl and click the column headers you want to merge, in the order you want them combined.
- Right-click → Merge Columns. OR Transform tab → Merge Columns.
- Choose a separator (Space, Comma, Colon, Semi-colon, or Custom).
- Enter a name for the new merged column.
- Click OK.

Important

When you merge columns, the original columns are removed. If you need to keep the originals, duplicate them first using Add Column → Duplicate Column before merging.

Chapter 5

Pivot & Unpivot

Pivot and Unpivot are two of the most powerful reshaping operations in Power Query. They allow you to change the structure of your data — turning rows into columns or columns into rows — without writing any formula.

5.1 Understanding the Concepts

Before applying these operations, it is important to understand the two formats of data:

Format	Description	Best Used For
Wide Format	Each category becomes its own column. Fewer rows, more columns.	Reports, dashboards, cross-tab summaries
Long Format	Each category-value pair occupies its own row. More rows, fewer columns.	Data analysis, charts, PivotTables, Power BI visuals

Pivot converts Long format to Wide format. Unpivot converts Wide format to Long format.

5.2 Unpivot — Wide to Long

Unpivot is the operation you will use most frequently. Raw data often comes in a wide format (months or regions as column headers). For analysis, Power BI and PivotTables require data in long format.

Example

You receive a quarterly sales report that looks like this:

Salesperson	Q1 Sales	Q2 Sales	Q3 Sales	Q4 Sales
Alice	50,000	42,000	48,000	55,000
Bob	38,000	41,000	36,000	43,000

Each quarter is a separate column. To build a chart or use a slicer for quarter, you need this in long format:

Salesperson	Quarter	Sales
Alice	Q1 Sales	50,000
Alice	Q2 Sales	42,000
...	...	...

Steps to Unpivot:

- In the Power Query Editor, select the columns you want to unpivot (the quarterly columns in our example).
- Right-click the selected columns → Unpivot Columns. OR Transform tab → Unpivot Columns.
- Power Query creates two new columns: Attribute (the column names) and Value (the values).
- Rename 'Attribute' to 'Quarter' and 'Value' to 'Sales'.

**Pro Tip — Unpivot Other Columns**

Instead of selecting all the quarter columns, try selecting the fixed columns (like Salesperson) and choosing 'Unpivot Other Columns'. This way, even if new quarter columns are added in future, they will be automatically unpivoted.

5.3 Pivot — Long to Wide

Pivot converts data from long format to wide format. You take a column of category values and spread each unique value into its own column.

Steps to Pivot:

- Select the column whose unique values you want to become new column headers (e.g., 'Quarter').
- Go to Transform tab → Pivot Column.
- In the Pivot Column dialog, select the Values Column (e.g., 'Sales').
- Choose an Aggregate Function (Sum, Average, Count, etc.).
- Click OK.

Chapter 6

Adding Custom Columns

Power Query allows you to create new columns based on calculations, conditions, or combinations of existing columns — all without leaving the Power Query Editor. This chapter covers the three main ways to add columns.

6.1 Custom Column — Write Your Own Formula

A Custom Column lets you write your own M formula to create a new column based on any logic you define.

How to add a Custom Column:

- Go to the Add Column tab → Custom Column.
- A dialog opens. Enter a name for the new column.
- Write the formula in the formula box. Use the column list on the right to insert column names.
- Click OK.



Example — Calculating Total Price

You have two columns: 'Unit Price' and 'Quantity'.
You want to create a new column 'Total Price' = Unit Price × Quantity.

In the Custom Column dialog:
New Column Name: Total Price
Formula: = [Unit Price] * [Quantity]

Note: Column names in M formulas are enclosed in square brackets [].

// Custom Column formula examples:

// Calculate profit margin:
= ([Selling Price] - [Cost Price]) / [Selling Price]

// Concatenate first and last name:
= [First Name] & " " & [Last Name]

// Calculate age from birth year (assuming current year is stored in a column):
= 2025 - [Birth Year]

Figure 6.2 — Sample Custom Column formulas in M language

6.2 Conditional Column — If/Else Logic

A Conditional Column creates a new column based on if-then-else logic — similar to an IF formula in Excel, but built through a point-and-click interface with no code required.

Example

You have a 'Score' column with student marks out of 100.
You want to create a 'Grade' column with the following rules:
Score >= 90 → 'A'
Score >= 75 → 'B'
Score >= 60 → 'C'
Otherwise → 'D'

Steps:

- Go to Add Column tab → Conditional Column.
- Enter the name of the new column (e.g., 'Grade').
- For each condition, click Add Clause and set:
 - Column Name: Score
 - Operator: is greater than or equal to
 - Value: 90
 - Output: A
- Continue adding clauses for B, C. Set the 'Otherwise' output to D.
- Click OK.

6.3 Column From Examples

Column From Examples is one of the most intuitive features in Power Query. You show Power Query one or two examples of what you want the new column to look like, and Power Query figures out the pattern automatically.

Example

You have a 'Full Name' column with values like 'Alice Sharma', 'Bob Mehta'.
You want to extract just the first name.

In the first row, you type 'Alice'. Power Query recognizes the pattern and fills in 'Bob' for the next row.
If the suggestion is correct, you accept it and click OK.

Steps:

- Go to Add Column tab → Column From Examples → From All Columns.
- A new column appears with empty cells. Type your example value in the first row.
- Power Query shows a suggested formula and fills in the remaining values.
- If the suggestion is correct, click OK. If not, type another example to guide it further.

Chapter 7

Applied Steps & Query Dependencies

One of Power Query's most important features is that it records every single transformation you apply — and stores them as a list of steps. This chapter explains how Applied Steps work and how queries can depend on each other.

7.1 What Are Applied Steps?

Every time you perform an action in Power Query — removing a column, changing a data type, filtering rows — it is recorded as an Applied Step in the Query Settings panel on the right.

Applied Steps

A sequential, ordered list of all the transformation actions performed on a query. Power Query executes these steps from top to bottom every time the query is refreshed.

Think of Applied Steps like a recipe — it tells Power Query exactly what to do to the raw data, in what order, to produce the final result.

7.2 Managing Applied Steps

Action	How to Do It
View data at any step	Click the step name. The data preview updates to show data at that point.
Delete a step	Click the ✕ icon to the left of the step name.
Edit a step	Click the gear ⚙ icon to the right of the step name. The original dialog reopens.
Rename a step	Right-click the step → Rename. Type a meaningful name.
Reorder steps	Right-click → Move Up or Move Down.
Add a step in between	Click the step after which you want to insert. Make your change — it inserts automatically.

Important Warning

Deleting or reordering steps can break the steps that come after them. For example, if Step 3 renames a column, and Step 5 uses that new column name — deleting Step 3 will cause Step 5 to fail because the column name no longer exists.

Always review downstream steps when modifying earlier steps.

7.3 The M Formula Language

Behind every Applied Step is a line of code written in Power Query's native language called M (also called Power Query Formula Language). You can see it in the Formula Bar at the top of the editor.

You do not need to write M code manually as a beginner — the interface creates it automatically. However, understanding the basic structure helps you debug errors and make small edits when needed.

```
// M formula automatically generated when you remove a column:
= Table.RemoveColumns(#"Changed Type",{"Customer Code", "Internal ID"})

// M formula for renaming a column:
= Table.RenameColumns(#"Removed Columns",{"cust_id", "Customer ID"})

// M formula for filtering rows (removing blanks):
= Table.SelectRows(#"Renamed Columns", each [Amount] <> null)
```

Figure 7.2 — M formulas generated automatically by Power Query transformations



Note

Each step references the previous step by name (the name is shown in quotes with # prefix). This is how steps are chained together. The first step is always named 'Source'.

7.4 Query Dependencies

As you work with Power Query, you can create multiple queries that build on each other. This is called a query dependency.

Query Dependency

When one query uses the output of another query as its source. The dependent query will automatically update whenever the source query is refreshed or modified.

Common use cases for dependent queries:

- **Reference Query:** Creates a new query that is a copy of an existing query. Changes to the original are reflected in the reference.
- **Appended Query:** Combines (stacks) rows from two or more queries into one.
- **Merged Query:** Joins two queries side-by-side using a common key (similar to SQL JOIN or Excel VLOOKUP).

How to view query dependencies:

- Go to View tab → Query Dependencies.
- A diagram appears showing all queries and the arrows indicating which queries feed into others.

7.5 Loading Data Back to Excel

Once your transformations are complete, you need to load the cleaned data back into Excel (or Power BI).

Options when you click Close & Load:

Load Option	What It Does
Close & Load	Loads data into a new Excel sheet as a formatted Table.
Close & Load To...	Opens a dialog to choose: Table, PivotTable, PivotChart, or Connection Only (for data model use).
Connection Only	Does not load data into a sheet. The query exists and can be used in Power Pivot or merged with other queries.

Chapter 8

Practice Exercise

This chapter provides a comprehensive, step-by-step practice exercise that covers all the key concepts taught in this module. You will work with a realistic raw dataset and apply all the transformations covered — from cleaning to reshaping to adding custom logic.

8.1 Dataset Description

For this exercise, you will use a simulated employee performance dataset exported from an HR system. The raw file has the following characteristics:

- Format: CSV file
- Name: HR_Performance_Raw.csv
- Rows: approximately 200 employee records
- Known issues: Extra header rows, blank rows, missing values, inconsistent date formats, combined name column, wide-format quarterly ratings

The file has the following columns in raw form:

Column Name (Raw)	Example Value	Issue to Fix
emp_id	EMP-1001	None — keep as is
full_nm	Alice Sharma	Split into First Name and Last Name
dept	HR	Rename to Department
join_dt	15-03-2020	Change to Date type; rename to Joining Date
salary	65000	Change to Whole Number; rename to Salary
Q1_rating	4.2	Part of wide-format data — Unpivot later
Q2_rating	3.8	Part of wide-format data — Unpivot later
Q3_rating	null	Part of wide-format data; null = not yet rated
Q4_rating	4.5	Part of wide-format data — Unpivot later

internal_code	SYS-XK-9	Not needed — remove this column
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8.2 Step-by-Step Instructions

Follow these steps exactly in the order given. Each step refers to the concept covered in a specific chapter.

Step 1 — Import the CSV File

- Open Microsoft Excel. Go to Data tab → Get Data → From File → From Text/CSV.
- Select HR_Performance_Raw.csv. In the preview, click Transform Data.
- The Power Query Editor opens.

Step 2 — Remove the Top 2 Rows (Chapter 2)

- The first two rows contain report metadata, not data.
- Home tab → Remove Rows → Remove Top Rows → enter 2. Click OK.
- The first row should now be the actual column headers.
- Click Use First Row as Headers if the column names are still in the first data row.

Step 3 — Remove Blank Rows (Chapter 2)

- Home tab → Remove Rows → Remove Blank Rows.

Step 4 — Remove the 'internal_code' Column (Chapter 2)

- Right-click the internal_code column header → Remove Columns.

Step 5 — Rename Columns (Chapter 2)

Rename the following columns by double-clicking each header:

Original Name	Rename To
emp_id	Employee ID
dept	Department
join_dt	Joining Date
salary	Salary

Step 6 — Change Data Types (Chapter 2)

- Joining Date column: Click the data type icon → select Date.
- Salary column: Click the data type icon → select Whole Number.
- Q1_rating, Q2_rating, Q3_rating, Q4_rating: Change each to Decimal Number.

Step 7 — Split 'full_nm' Column (Chapter 4)

- Select the full_nm column.
- Transform tab → Split Column → By Delimiter → choose Space → split at left-most delimiter.
- Rename the two resulting columns: full_nm.1 → First Name, full_nm.2 → Last Name.

Step 8 — Handle Missing Values in Q3_rating (Chapter 3)

- Right-click Q3_rating → Replace Values.
- Leave 'Value to Find' blank. In 'Replace With', enter 0. Click OK.
- (Interpretation: An employee not yet rated in Q3 receives 0 as a placeholder.)

Step 9 — Add a Custom Column for Average Rating (Chapter 6)

- Add Column tab → Custom Column.
- Name the column: Avg Rating.
- Formula: $= ([Q1_rating] + [Q2_rating] + [Q3_rating] + [Q4_rating]) / 4$
- Click OK.

Step 10 — Add a Conditional Column for Performance Category (Chapter 6)

- Add Column tab → Conditional Column.
- Name the column: Performance Category.
- Add clauses:
 - Avg Rating ≥ 4.5 → 'Outstanding'
 - Avg Rating ≥ 3.5 → 'Meets Expectations'
 - Otherwise → 'Needs Improvement'
- Click OK.

Step 11 — Unpivot the Quarterly Ratings (Chapter 5)

- Hold Ctrl and select all four rating columns: Q1_rating, Q2_rating, Q3_rating, Q4_rating.
- Transform tab → Unpivot Columns.
- Two new columns appear: Attribute and Value.
- Rename Attribute → Quarter, Value → Rating.

Step 12 — Review Applied Steps (Chapter 7)

- In the Query Settings panel on the right, review all your Applied Steps.
- Click each step to verify the data looks correct at that point.
- Rename any automatically-named steps to more descriptive names (e.g., 'Removed internal_code').

Step 13 — Load Data to Excel (Chapter 7)

- Home tab → Close & Load → Close & Load To...
- Select Table. Choose a destination cell (e.g., cell A1 in a new sheet).
- Click OK. Your cleaned and transformed data loads into Excel.

Expected Final Output

Your final table should have the following columns:

Employee ID | First Name | Last Name | Department | Joining Date | Salary | Avg Rating | Performance
Category | Quarter | Rating

Each employee will have multiple rows — one per quarter. The data is clean, correctly typed, and analysis-ready.

Module Summary

In this module, you have covered the complete Power Query workflow — from importing raw data to loading clean, structured data into Excel. Here is a recap of the key concepts:

- Power Query is a no-code data transformation tool built into Excel and Power BI. It records all transformations as reproducible, refreshable steps.
- The Power Query Editor has five main areas: Ribbon, Query Pane, Data Preview, Applied Steps, and Formula Bar.
- Data Cleaning covers removing unwanted rows and columns, renaming columns for clarity, and assigning the correct data types.
- Missing Values (null) can be handled by removing rows, replacing with fixed values, or using Fill Down / Fill Up.
- Columns can be split into multiple columns using delimiters or character positions, and multiple columns can be merged with a custom separator.
- Pivot converts long-format data to wide-format; Unpivot does the reverse — and is the more commonly needed operation in analysis workflows.
- Custom Columns, Conditional Columns, and Column From Examples allow you to calculate and derive new columns from existing data.
- Applied Steps are a transparent, editable log of all transformations — making it easy to audit, fix, or update your data pipeline.
- Query Dependencies allow queries to reference and build on each other, enabling modular and maintainable data workflows.

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